U-FNO tests

	Regular FNO	FNO with width = 33	U-FNO	FNO-CNN
Number of parameters	**3,149,090**	3,348,974	3,274,914	3,204,578
Total training time (minutes)	**371.0**	503.8	1343.0	875.1
Final loss	9.9875E-07	8.9477E-07	9.7017E-08	**8.9862E-08**
Mean L2 error	1.1864%	1.1010%	0.4185%	**0.3929%**

Model	32_32_32	32_32_64	32_32_64_bilinear	32_64_128
Number of parameters	3,280,546	3,391,266	3,391,266	3,631,074
Training duration	1302.8	1462.6	1566.4	1760
Final loss	9.2112E-08	8.7037E-08	1.1516E-07	8.3210E-08
Average L2 error	0.4017%	0.3820%	0.4250%	0.3748%

This batch of tests is to determine the best configuration of the Unet inside of the FNO blocks.

The previous (baseline UFNO) model has a configuration for the **channel width of 16 – 32 – 64 (at the bottleneck).** In this test, we also test other configuration , such as 32-32-64.

I also tested a different upsampling method, from regular nearest repeat (in regular baseline UFNO) to bilinear interpolation.

Results:

- Surprisingly, even a configuration of 32-32-32 works better than 16-32-64. The model has a width of 32, so the first 16 conv layer compressed the channel width. Training time is shorter despite having more parameters.
- Best perfoming configuration is 32-64-128 obviously. However, it has significantly more parameters and training time is also significantly longer as well.
- Bilinear interpolation for the upsampling part made the model perform much worse, even worse than the original UFNO model. Training time is longer as well.

Conclusion:

1. (not entirely sure) The first conv layer should have an input at least equal to the model width. Compressing from 32 to 16 in the base model is not very logical and impact the performance slightly.

2. Nearest neighbor repeat is better than bilinear interpolation for upsampling in the UNET in every aspect.
3. Optimal setting should be 32-64-128, but if the training time is too long or there are too many parameters, 32-32-64 is also good.

Zero shot super resolution

U-FNO	0.4185%	3.5545%
U-FNO with dilation		5.1448%
U-FNO with down-sampled input		**1.8111%**

Model	32_32_32	32_32_64	32_32_64_bilinear	32_64_128
64x64	0.4017%	0.3820%	0.4250%	0.3748%
256x256	4.6074%	4.5647%	4.7275%	4.7431%
256x256 dilation	2.4021%	3.1285%	1.5748%	4.7266%
256x256 input resizing	2.1334%	2.1472%	2.1538%	2.1591%

- Resizing input before works consistently with U-FNO. Dilation behaves much weirder.
- It also seems like, the better the 64x64 result is, the worse the 256x256 result is.

Adding CNN/ UNET to the lifting block tests

Model	CNN-input concat	CwC input	Input-CNN (reg.FNO)	Unet-64	Unet 128
Number of parameters	3,159,682	3,214,146	3,158,658	3,267,362	3,631,970
Training duration	584.4	1164	578.3	1032.3	1146.4
Final loss	2.3584E-07	6.4734E-08	2.3180E-07	1.5425E-07	1.5473E-07
Average L2 error	0.5833%	0.3449%	0.5758%	0.5062%	0.5043%

CNN-input concat: the CNN branch output is concatenated with the input features (32 + 3 = 35), then feed through the lifting layer

CwC input: regular CNN branch in the lifting block in additional to a regular FNO-CNN network

Input-CNN: regular CNN branch in the lifting block in additional to a regular FNO network

Unet-64: a Unet with max = 64 channel at the bottleneck, concatenated with the input channels, then lifting

Unet-128: a Unet with max = 128 channel at the bottleneck, concatenated with the input channels, then lifting

Conclusion:

- Adding the CNN branch output and the lifting layer output seems to work better than concatenating
- Unet larger size doesn't make much difference here
- Unet preprocessing the input seems to be more effective, but with obviously more parameters used

One shot super resolution for CwC model:

64x64: 0.3449%

256x256: 5.9442% (regular)

256x256: 1.4833% (with dilation both in the lifting and the FNO blocks)

256x256: 2.4117% (with resampling input in the lifting block and dilation in the FNO block)

256x256: 3.0055% (with resampling for both)

256x256: 1.9888% (dilation lifting + resampling FNO block)

Conclusion: dilation still works best wherever there is CNN

Mixing different lifting block options with FNO block options

Model	CNN input U-FNO	UNET input CNN	UNET input U-FNO	CNN input CNN
Number of parameters	3,284,482	3,322,850	3,393,186	3,214,146
Training duration	1258	1766.1	2131.2	1164
Final loss	7.6996E-08	5.7729E-08	7.8146E-08	6.4734E-08
Average L2 error	0.3689%	0.3203%	0.3715%	0.3449%

Zero shot super resolution

Model	CNN input U-FNO	UNET input CNN	UNET input U-FNO	CNN input CNN
64x64	0.3689%	0.3203%	0.3715%	0.3449%
256x256	4.5481%	5.2127%	4.7227%	5.9442%
dilation-dilation	2.0829%	**1.5236%**	**1.6654%**	**1.4833%**
dilation-resampling	**1.6255%**	1.9969%	1.7066%	1.9888%
resampling-diation	2.6401%	2.2387%	2.1006%	2.4117%
resampling-resampling	2.8621%	2.8427%	2.4092%	3.0055%